

Discussion 2: Symbolic Approach vs Neural-Network-Based Approach

Discussion Topic:

Regarding symbolic approaches and connectionist neural-network-based approaches, do you think one has an advantage over the other regarding a particular application? Justify with an example.

My Post:

Hello Class

In the field of artificial intelligence, the symbolic approach and the connectionist neural-network-based approach are very different approaches to using AI for problem-solving. The symbolic approach represents knowledge through explicit symbols, concepts, relationships, and predefined rules. This approach uses logical operations, such as deduction and induction, to process and manipulate those symbols to reach conclusions or predictions. This approach is best suited when a problem can be clearly defined through rules and the outcomes are deterministic. However, it requires humans to manually construct and maintain a large knowledge base, and its sets of rules can grow rapidly in number and complexity as the problem domain grows more complex (Colorado State University Global, n.d.; Newell, 1980). For some problems, the complexity can become insurmountable because every relevant concept, relationship, exception, and rule must be explicitly represented and manually programmed, which makes the system difficult to scale, update, and maintain.

On the other hand, the connectionist neural-network-based approach uses networks of interconnected nodes, or artificial neurons, connected by weighted edges, or artificial neural connections. Through learning algorithms such as gradient descent and backpropagation, the edge weights can be adjusted during the training phase of an Artificial Neural Network (ANN) model. Thus, instead of processing explicitly human-programmed rules, such as if-then-else statements, to generate an output, an ANN model generates outputs from learned statistical patterns directly extracted from data sets using techniques such as Supervised Learning (SL), in which the model learns from labeled examples containing known input-output pairs, and Unsupervised Learning (UL), in which the model identifies patterns, structures, clusters, or relationships in data without predefined labels or target outputs. ANN models can also be trained using Reinforcement Learning (RL), in which an agent learns through interactions with an environment and receives rewards or penalties based on the consequences of its actions, and Self-Play Reinforcement Learning (SPRL), in which an agent improves by repeatedly competing or cooperating against copies or earlier versions of itself rather than relying exclusively on human-generated training examples. Additionally, ANNs can be categorized based on the number of hidden layers, or artificial neuron layers, a specific network contains between its input and the output layer. For instance, Shallow Neural Networks (SNNs) are ANNs that generally contain one, and sometimes two, hidden layers. Deep Neural Networks (DNNs), on the other hand, contain multiple hidden layers and therefore can learn more abstract representations of their input data; they can acquire a 'deep' representation of a specific `world`. By adding multiple neuron layers, DNNs effectively create multiple levels of abstraction within their internal representations, giving them the ability to perform complex analyses such as image recognition, object detection, speech processing, and other tasks involving large quantities of complex or unstructured data (Colorado State University

Global, n.d.; LeCun et al., 2015). Generative AI models such as Large Language Models (LLMs) are transformer-based DNNs that learn statistical relationships among words or tokens from extremely large text data to generate text based on a text input (a prompt) using an attention mechanism that predicts the most probable next text token.

When comparing the symbolic approach and the connectionist neural-network-based approach, the connectionist neural-network-based approach seems to be superior and may be superior in many domains. However, this appearance of superiority is context-dependent, meaning that the most effective approach depends on the nature of the problem and on whether it can be approached deterministically through explicit rules or probabilistically through patterns learned from data. DNNs perform best in applications that require tasks to identify complex, nonlinear patterns within large datasets having a large amount of noisy or unstructured data. Symbolic-based models, on the other hand, performed best in applications where knowledge can be represented through explicit rules or be programmed, and where outcomes must meet clear constraints. Therefore, it can be argued that neither approach is superior and they are context-dependent. In other words, their advantages depend on the structure of the problem, the available data, the need for interpretability, and the degree to which the domain can be expressed through formal rules. For complex applications that require both perception and logical reasoning, a hybrid approach may provide the greatest overall advantage by combining the pattern-learning capabilities of neural networks with the explicit knowledge representation and reasoning capabilities of symbolic AI (Colorado State University Global, n.d.).

For example, in the autonomous-vehicle field, visual awareness or perception is essential for the vehicle to detect in real-time pedestrians, vehicles, lane markings, traffic signs, and obstacles and for responding quickly to changing road conditions. A connectionist neural-network-based approach brings significantly more advantages than a symbolic approach for such real-time tasks. DNNs can learn directly from large collections of images to recognize a large number of features or objects rather than, as is required by the symbolic approach, for engineers to define an explicit rule for every feature or object, viewing angle, lighting condition, or road configuration. For instance, Bojarski et al. (2016) trained a Convolutional Neural Network (CNN) to convert images from a vehicle's front-facing camera directly into steering commands. A CNN is a type of DNN that processes spatially structured data, such as images, by applying learnable filters across local sections of the input images. The filters are small matrices of weights, also called kernels, that slide across an image and calculate responses to local visual patterns. These weights are tuned during training using backpropagation together with a gradient-based approach such as Stochastic Gradient Descent (SGD). Using this method, Bojarski et al.'s system learned how to drive on highways and local roads, including roads without visible lane markings.

Although a symbolic approach would have a difficult time accounting for every possible visual variation of every feature and object, as each variation of an object should be defined by a rule. Nonetheless, symbolic AI remains advantageous for representing explicit rules such as traffic laws and safety constraints, for instance, reducing speed during hazardous weather conditions, such as heavy rain, snow, fog, or icy roads. Therefore, it can be argued that the strongest design for an autonomous vehicle may be a hybrid approach in which CNNs perform perception and pattern recognition and a symbolic reasoning component of the system makes driving decisions. Research into neuro-symbolic autonomous-driving systems has shown that neural representations and symbolic reasoning can be combined within the same decision-making architecture (Sun et al., 2021).

Moreover, recent research from AI laboratories suggests that the next stage of AI reasoning may include a mix of Large Reasoning Models (LRMs) and Neuro-Symbolic Models (NSM). LRMs are transformer-based DNNs designed to generate outputs using chains of thought processes. On the other hand, Neuro-symbolic models (NSMs) integrate a DNN approach with symbolic knowledge illustration and reasoning processes, such as rules, search algorithms, planners, or theorem provers (Yang et al., 2025). These knowledge and reasoning components allow the DNN component of the model to generate answers, plans, or hypotheses based on symbolic components. For example, Trieu et al. (2024) describe their AlphaGeometry model as an NSM.

In the near future, the direction of AI development may imply hybrid systems in which LRMs are augmented by symbolic reasoning components, birthing a new type of LRM that may be defined as a Large Symbolic Reasoning Model (LSRM). Furthermore, Yang et al. (2024), Garzes and Lamb (2023), Marcus (2020), Hassabis (2026), and others argue that a hybrid approach to AI development, such as a new type of LLM-Neuro-Symbolic AI, is a critical step to reach Artificial General Intelligence (AGI). Although this is speculative, the incoming OpenAI GPT-6 model may be a type of LSRM.

-Alex

References

- Bojarski, M., Del Testa, D., Dworakowski, D., Firner, B., Flepp, B., Goyal, P., Jackel, L. D., Monfort, M., Muller, U., Zhang, J., Zhang, X., Zhao, J., & Zieba, K. (2016). *End to end learning for self-driving cars*. arXiv. <https://doi.org/10.48550/arXiv.1604.07316>
- Colorado State University Global. (n.d.). *Lecture 2: Foundational theories and their application* [Course lecture]. Canvas.
- Garcez, A. d'Avila, & Lamb, L. C. (2023). Neurosymbolic AI: The 3rd wave. *Artificial Intelligence Review*, 56(11), 12387–12406. <https://doi.org/10.1007/s10462-023-10448-w>
- Hassabis, D. (2026, March 10). *From games to biology and beyond: 10 years of AlphaGo's impact*. Google DeepMind. <https://deepmind.google/blog/10-years-of-alphago/>
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444. <https://doi.org/10.1038/nature14539>
- Marcus, G. (2020). *The next decade in AI: Four steps towards robust artificial intelligence* [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2002.06177>
- Newell, A. (1980). Physical symbol systems. *Cognitive Science*, 4(2), 135–183. https://doi.org/10.1207/s15516709cog0402_2
- Sun, J., Sun, H., Han, T., & Zhou, B. (2021). Neuro-symbolic program search for autonomous driving decision module design. In J. Kober, F. Ramos, & C. Tomlin (Eds.), *Proceedings of the 2020*

Conference on Robot Learning (Vol. 155, pp. 21–30). Proceedings of Machine Learning Research.
<https://proceedings.mlr.press/v155/sun21a.html>

Trinh, T. H., Wu, Y., Le, Q. V., He, H., & Luong, T. (2024). Solving olympiad geometry without human demonstrations. *Nature*, 625(7995), 476–482. <https://doi.org/10.1038/s41586-023-06747-5>

Yang, X.-W., Shao, J.-J., Guo, L.-Z., Zhang, B.-W., Zhou, Z., Jia, L.-H., Dai, W.-Z., & Li, Y.-F. (2025). Neuro-symbolic artificial intelligence: Towards improving the reasoning abilities of large language models. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence* (pp. 10770–10778). International Joint Conferences on Artificial Intelligence Organization.
<https://doi.org/10.24963/ijcai.2025/1195>